

POSTGRESQL ANALYTICS REPORT – Retail Customer Behavior

1. Objective

Execute a structured SQL-based analysis on the *customer_shopping_behavior* dataset to extract business-critical KPIs across revenue, category mix, customer segments, geographies, and seasonality.

2. Dataset Structure (in PostgreSQL)

Table: **sales**

Rows: **3,900**

Columns include:

customer_id, age, gender, item_purchased, category, purchase_amount_usd, location, season

Dataset contains **one transaction per customer**.

3. KPI Outputs (from SQL queries)

3.1 Business Volume

- **Total revenue:** 233,081 USD
- **Total transactions:** 3,900
- **Unique customers:** 3,900

Interpretation: even distribution → standardized dataset.

3.2 Average Basket Value (ABV)

- **59.76 USD**

Core spending benchmark for all segments.

3.3 Category Performance

Category	Revenue (USD)
Clothing	104,264
Accessories	74,200
Footwear	36,093
Outerwear	18,524

Key takeaway: Clothing + Accessories = **76% of total revenue** → core assortment.

3.4 Customer Value

Each customer:

- **Spent 100 USD**
- **Performed 1 transaction**

Segmentation must be based on **attributes**, not monetary value.

3.5 Gender Insights

Gender	Transactions	Revenue	Avg Ticket
Male	2,652	157,890	59.54
Female	1,248	75,191	60.25

Takeaway:

Male = **68%** of revenue.

Female = *slightly higher ABV*.

3.6 Location Performance (Top 5)

State	Transactions	Revenue
Montana	96	5,784

Illinois	92	5,617
California	95	5,605
Idaho	93	5,587
Nevada	87	5,514

Revenue is **balanced across locations**, no extreme concentration.

3.7 Seasonality

Season	Revenue
Fall	60,018
Spring	58,679
Winter	58,607
Summer	55,777

Takeaway:

Fall = strongest season.
Summer = weakest.

4. Strategic Synthesis

- Double down on Clothing + Accessories
- Strengthen Fall campaigns
- Leverage male segment volume; upsell premium female segment
- Deploy geo-targeted actions for low-performing regions
- Use attribute-based segmentation (gender, location, season, category)